# Sterling Hall

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Machine learning engineer with experience in LLM fine-tuning, biosignal time-series modeling, privacy-preserving ML, and ML-driven threat detection

## Education

**Princeton University**, MS in Electrical and Computer Engineering — Sep 2025 – May 2026

- **Relevant Coursework:** Reinforcement Learning Theory, Computational Biology, Molecular Modeling Methods, Computer Security

**Princeton University**, BSE in Electrical and Computer Engineering — Sep 2021 – May 2025

- **Thesis:** GRU-SFL: A Private, Secure, Efficient Machine Learning Architecture for Healthcare; designed a split federated learning architecture with GRU models for privacy-preserving healthcare ML

## Skills

**Languages:** Python, TypeScript, SQL
**Machine Learning:** PyTorch, LLM fine-tuning (SFT, QLoRA), LSTM and GRU networks, split/federated learning, reinforcement learning, dataset curation and annotation
**Engineering:** React, React Native, NestJS, Prisma, Supabase, AWS Cognito, MQTT, Git

## Experience

**Security Analyst**, Aegis AI – New York, NY — Sep 2026 – Present

- Annotate 5,000+ emails per week with 99.7% accuracy, building labeled training datasets for ML-based email threat detection models
- Issue indicator of compromise (IOC) rulings on suspicious and malicious domains to support threat intelligence and email security
- Integrate identified threat patterns into rules-based detection systems, complementing ML models with deterministic detection logic

**Co-Founder, COO**, eir – Princeton, NJ — Dec 2023 – May 2025

- Founded an AI sports analytics company for female athletes; raised pre-seed funding from multiple sources and established a partnership with USA Rowing to validate the platform through athlete trials
- Fine-tuned an LLM with Supervised Fine-Tuning (SFT) and QLoRA on a curated dataset, and built an LSTM neural network to predict menstrual cycle symptoms and performance trends from biosignal data
- Led design and rapid prototyping of a chest-worn wearable based on Polar H10; trials showed increased comfort and equivalent accuracy to established market alternatives
- Built mobile and web applications using React Native, TypeScript, and Supabase
- Selected for Keller Center’s eLab Summer Accelerator program (Summer 2024)

**Full-Stack Developer**, Substrate Industries – San Francisco, CA — Jun 2025 – Aug 2025

- Led 0-to-1 development of two production applications, credit modules, and three customizable demo platforms helping U.S. manufacturers access $15M+ in flexible financing across 8 early customers
- Designed Prisma backend schemas and NestJS endpoints for secure inter-app data sharing, and built React UI components across the platform

**Software Engineering Research Intern**, CoNSeRT Lab – Athens, Greece — Jun 2023 – Aug 2023

- Engineered a lightweight MQTT messaging network for a resource-constrained AR environment, creating a secure subscription-based notification system for critical vitals thresholds with 250ms end-to-end latency
- Developed an Android application in Kotlin using Vuzix HUD packages for simultaneous biometrics monitoring and rescue operations

## Leadership

**Teaching Assistant**, ECE 115, 203, 206 — Sep 2023 – May 2026

- Led labs and tutorials in circuit design, computer logic design, and drone robotics control for 75+ students across three courses
- Developed supplementary materials and evaluated autonomous navigation projects and circuit implementations

**Officer**, Engineers Without Borders — Sep 2021 – May 2025

- Led a 7-member team designing and simulating a pipeline addition to serve 1,500+ community members using EPANET, with professional advisors and MIWASCO